

Ahmed Gamal Badawy

412200089

Gimy Crypto Tool Documentation

Code on my Github:

<https://github.com/gimmy554/Crypto-Tool/blob/main/GimyCryptoTool.py>

Overview

Gimy Crypto Tool is a Python-based graphical application built with Tkinter that provides a user-friendly interface for performing various cryptographic operations. It supports symmetric and asymmetric encryption, classical ciphers, hashing, and digital signatures, making it a versatile tool for learning and experimenting with cryptography.

The tool is designed for educational purposes and offers features such as key generation, file import/export, and a help section to guide users through its functionalities.

Features

- **Symmetric Encryption:** Supports AES and DES algorithms for encrypting and decrypting messages.
- **Asymmetric Encryption:** Implements RSA encryption and decryption using public/private key pairs.
- **Classical Ciphers:** Includes Caesar and Vigenère ciphers for educational purposes.
- **Hashing:** Supports MD5, SHA-1, SHA-256, and SHA-512 hashing algorithms.
- **Digital Signatures:** Allows signing and verifying messages using RSA-based signatures.
- **File Operations:** Load messages from files and save results to files.

- **Help Section:** Provides detailed instructions for using each feature.
- **Status Bar:** Displays feedback on operations (e.g., success, errors).

Installation

1. **Clone or Download the Project:** Download the project files or clone the repository to your local machine.

Set Up a Virtual Environment (Optional but recommended):

```
python -m venv venv
```

2. `source venv/bin/activate` # On Windows: `venv\Scripts\activate`
3. **Install Dependencies:** Install the required Python libraries using pip:
`pip install pycryptodome cryptography`
4. **Run the Application:** Execute the main script to launch the application:
`python GimyCryptoTool2.py`

Usage

Launching the Application

Run the script `GimyCryptoTool2.py` to open the graphical interface. The window is divided into a navigation menu on the left and a main content area on the right, with a status bar at the bottom.

Navigation

Use the buttons on the left to switch between tabs:

- **Symmetric:** For AES/DES encryption and decryption.
- **Asymmetric:** For RSA encryption and decryption.
- **Classical:** For applying Caesar and Vigenère ciphers.
- **Hashing:** For generating hashes using various algorithms.

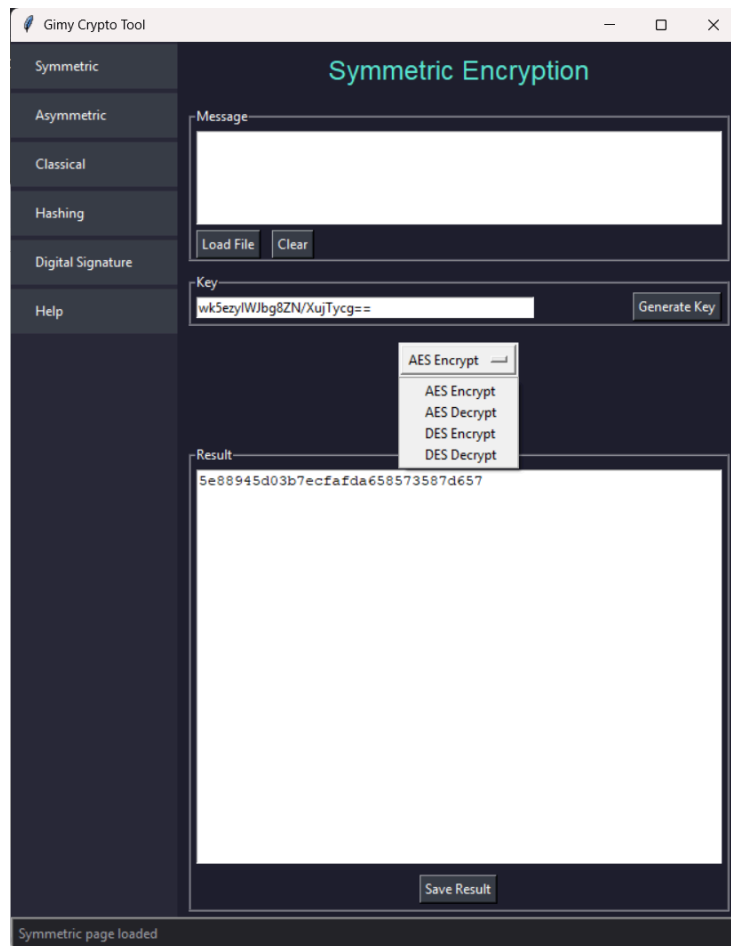
- **Digital Signature:** For signing and verifying messages.
- **Help:** Opens a help window with detailed instructions.

General Features

- **Load File:** Import a message from a .txt file into the input area.
- **Clear:** Clear the input area.
- **Save Result:** Export the output to a .txt file.
- **Status Bar:** Check the bottom status bar for feedback on operations.

Tab-Specific Usage

1. Symmetric Encryption (AES/DES)



- **Purpose:** Encrypt or decrypt messages using symmetric algorithms.
- **Steps:**
 1. Enter your message in the input area or load it from a file.
 2. Generate a key using the "Generate Key" button (AES: 16 bytes, DES: 8 bytes) or input your own base64-encoded key.
 3. Select an operation: AES Encrypt, AES Decrypt, DES Encrypt, or DES Decrypt.
 4. Click "Execute" to process the message.
 5. View the result in the output area and save it if needed.
- **Note:** DES is considered insecure for modern use; prefer AES for better security.

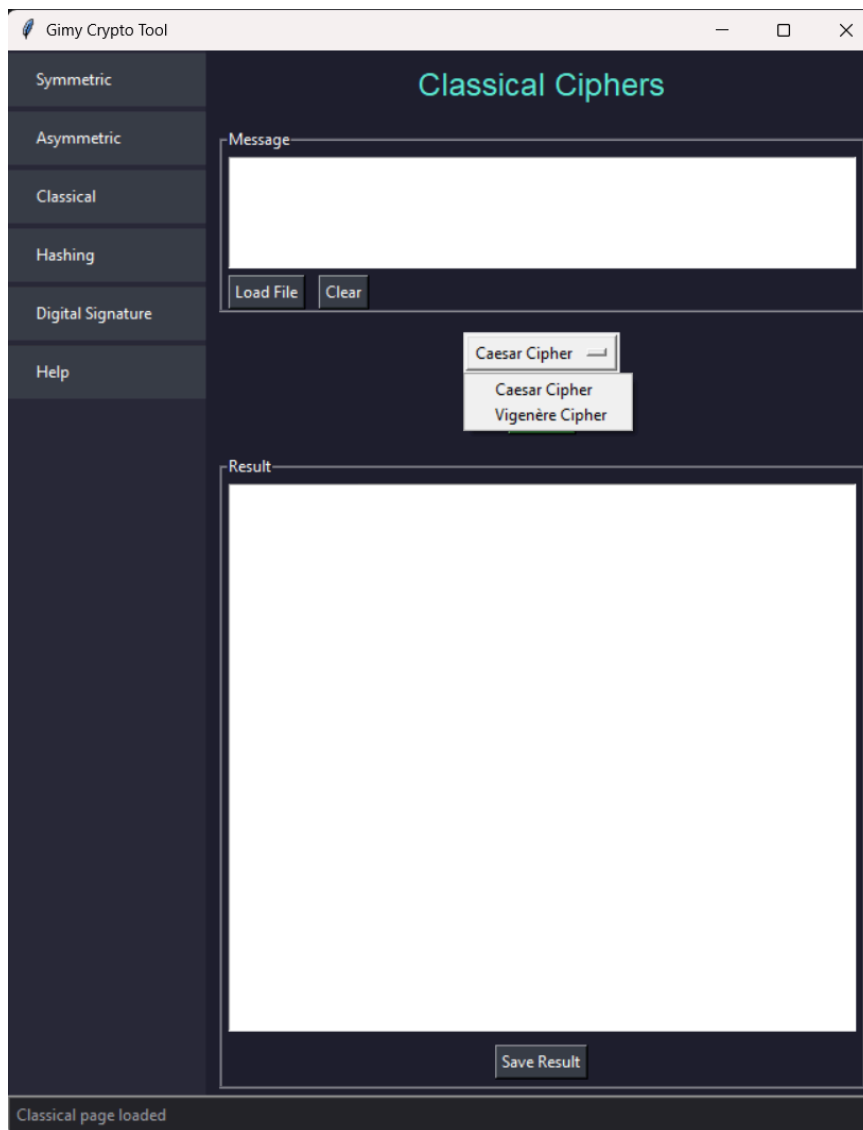
2. Asymmetric Encryption (RSA)

The screenshot shows the 'Gimy Crypto Tool' window with the 'Asymmetric Encryption (RSA)' tab selected. The interface includes a sidebar with navigation options: Symmetric, Asymmetric (selected), Classical, Hashing, Digital Signature, and Help. The main area contains a 'Message' input field with 'Load File' and 'Clear' buttons. Below this is the 'RSA Keys' section, which displays a 'Public Key' and a 'Private Key' in base64 format, with a 'Generate Keys' button. A dropdown menu is open, showing 'RSA Encrypt' (selected), 'RSA Encrypt', and 'RSA Decrypt'. At the bottom is a 'Result' output area with a 'Save Result' button. A status bar at the very bottom indicates 'Asymmetric page loaded'.

- **Purpose:** Encrypt or decrypt messages using RSA.

- **Steps:**
 1. Enter your message in the input area.
 2. Generate a key pair using the "Generate Keys" button.
 3. Select "RSA Encrypt" (uses public key) or "RSA Decrypt" (uses private key).
 4. Click "Execute" to process the message.
 5. View the result in the output area.
- **Note:** RSA is suitable for small messages due to size limitations.

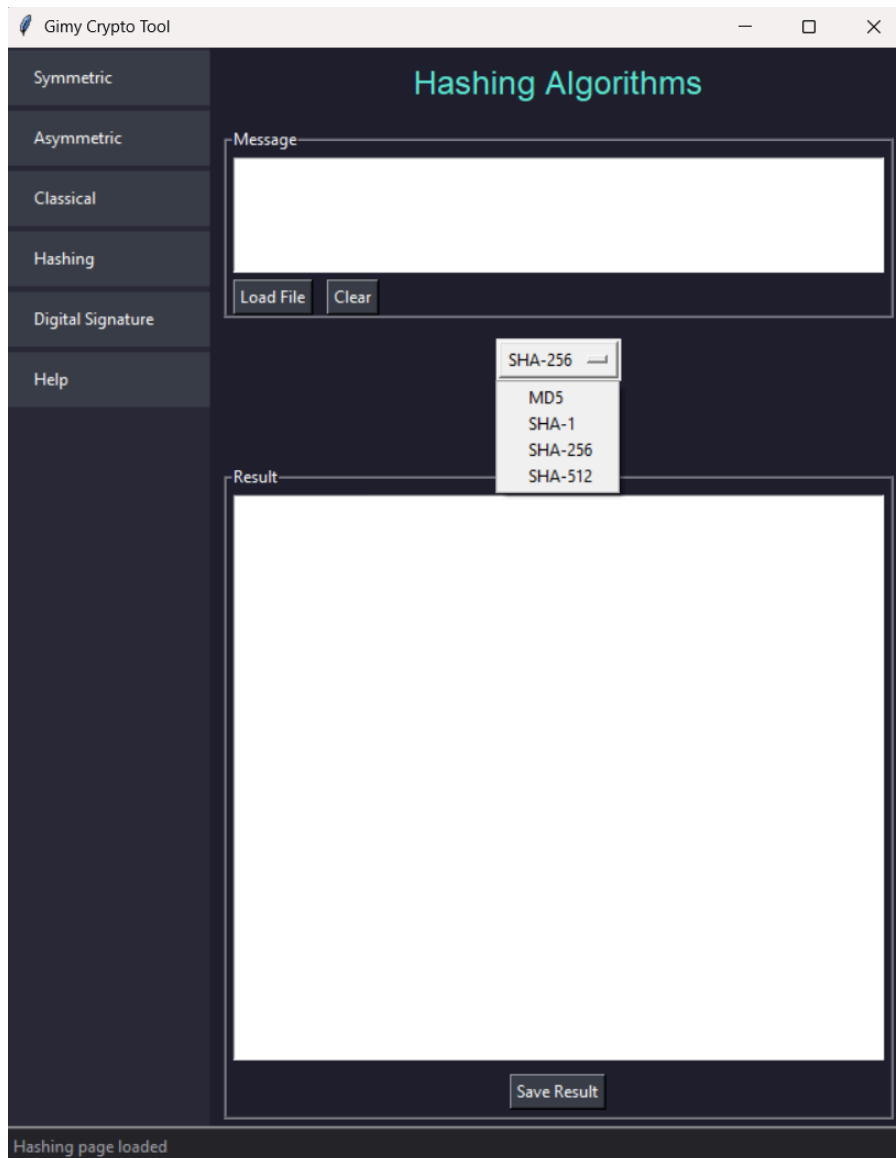
3. Classical Ciphers



- **Purpose:** Apply classical ciphers for educational purposes.
- **Steps:**

1. Enter your message in the input area.
 2. Select either "Caesar Cipher" (fixed shift of 3) or "Vigenère Cipher" (uses keyword "crypto").
 3. Click "Execute" to apply the cipher.
 4. View the result in the output area.
- **Note:** These ciphers are not secure for modern use.

4. Hashing Algorithms



- **Purpose:** Generate hashes of messages.
- **Steps:**
 1. Enter your message in the input area.

2. Select a hashing algorithm (MD5, SHA-1, SHA-256, or SHA-512).
 3. Click "Hash It!" to generate the hash.
 4. View the hash in the output area.
- **Note:** Use SHA-256 or SHA-512 for better security; MD5 and SHA-1 are considered weak.

5. Digital Signature

Gimy Crypto Tool

Digital Signature

Message

Load File Clear

RSA Keys

Public Key:

MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAwMCNrehYFANCb/
9J3ME2
B3kj1FwUcu/BdhMBduhKkohn3vTqFWUKdIubaoT1fXvytBtVOfbWD+NCMG

Private Key:

MIIEvAIBADANBgkqhkiG9w0BAQEFAASCBywggSiAgEAAoIBAQAawI2t6F
gUA0Jv
/0ncwTYHeSOIXBRy78F2EwF26EqSiGfe9OoVZQp0i5tqhPV9e/K0G1U59t

Generate Keys

Signature (for verification)

grererh

Sign

Sign
Verify

Result

Save Result

Digital Signature page loaded

- **Purpose:** Sign messages or verify signatures using RSA.
- **Steps:**
 1. Enter your message in the input area.
 2. Generate a key pair using the "Generate Keys" button.

3. To sign: Select "Sign," click "Execute," and the signature will appear in the output.
 4. To verify: Paste the signature in the "Signature (for verification)" area, select "Verify," and click "Execute."
 5. The result will indicate if the signature is valid.
- **Note:** Keep your private key secure, as it's used for signing.

6. Help

- **Purpose:** Access instructions for using the tool.
- **Steps:**
 1. Click the "Help" button in the navigation menu.
 2. A new window will open with detailed instructions for each tab.
 3. Scroll through the instructions and click "Close" when done.

